

D & f - block Elements

Introduction:- [D-block Elements]

→ **Transition Element**: - According to IUPAC, transition metals are defined as metals which have incomplete 'd' subshell either in neutral atom or in their ions [known O.S].

- Partially filled 'd' $\rightarrow (d^1 - d^9)$
- Zn, Cd, Hg $\rightarrow$ Transition elements ~~X~~
Reason $\rightarrow d^0$ configuration

- Ag Ag^+ Ag^{2+}
 $4d^{10}5s^1$ $4d^{10}5s^0$ $4d^95s^0$

→ Ag is trans-element bcz, it has partially filled 'd' in its +2 o.s.

D-block General electronic config:

$$(n-1)d^{1-10} ns^0 1/2$$

$n s^0 \rightarrow$ only in 'Pd'

$ns' \rightarrow \text{exceptional}$

$n\ell^2 \rightarrow$ Normal configuration

Some Important Examples

- Cr $\rightarrow$ [Ar] $3d^5 4s^1$
- Cu $\rightarrow$ [Ar] $3d^{10} 4s^1$
- Nb $\rightarrow$ [Kr] ~~$4d^3 5s^2$~~ $4d^4 5s^1$
- Mo $\rightarrow$ [Kr] $4d^5 5s^1$
- Ru $\rightarrow$ [Kr] $4d^7 5s^1$
- Rh $\rightarrow$ [Kr] $4d^8 5s^1$
- Pd $\rightarrow$ [Kr] $4d^{10} 5s^0$
- Ag $\rightarrow$ [Kr] $4d^{10} 5s^1$

f-block General electronic config.

$$ns^2 (n-2)f^{1-14} (n-1)d^{0-11}$$

$$(n-1)d^2 \rightarrow \text{only } 100 \text{ Th } (90)$$
$$(n-1)d^0 \rightarrow (\text{normal})$$

$(n-1)d' \rightarrow \text{exceptional}$

Atomic size

- 3rd group $\rightarrow 3d < 4d < 5d$ [No L.C]
 eg $\rightarrow (Sc < Y < La)$
- (4th - 12th) $\rightarrow 3d < 4d \approx 5d$ [due to group L.C]
 (Co, Ni, Cu + due to poor σ of 4f e⁻)
 E.g $\rightarrow Ti < Zr \approx Hf$

Metallic Radius order

(3d series)

(3d series) 137
 $\text{Sc} > \text{Ti} > \text{V} > \text{Cr} < \text{Mn} > \text{Fe} \approx \text{Co} \approx \text{Ni} < \text{Cu} < \text{Zn}$
 137

* $Sc > Zn$

★ Mn = Zn

Ionization Enthalpy

⑥ $L \rightarrow R$ → Min. for the first element of each series & Max for the last element in each series.

$\begin{pmatrix} T \\ \downarrow \\ B \end{pmatrix} \rightarrow$ except 3rd group i.e. for the 5d series element is max^m in each group due to L.C.

SC > Y > LA → NO L.C

→ Final order for 3d series

vinita thi nahi kon Jane
 $Sc < V < Cr < Ti < Mn < Ni < Cu < Co < Fe < Zn$
 suno kamal Man kya uka bis

Imp points

- $\bullet \text{Cr} < \text{Mn} (\text{I.E}_1)$ $\bullet \text{Mn} < \text{Fe} (\text{I.E}_1)$
 $3d^5 4s^1$ $3d^5 4s^2$ $3d^5 4s^2$ $3d^6 4s^2$
 $\bullet \text{Cr} > \text{Mn} (\text{I.E}_2)$ $\bullet \text{Mn} < \text{Fe} (\text{I.E}_2)$
 $3d^5$ $3d^5 4s^1$ $3d^5 4s^1$ $3d^6 4s^1$
 $\bullet \text{Cr} < \text{Mn} (\text{I.E}_3)$ $\bullet \text{Mn} > \text{Fe} (\text{I.E}_3)$
 $3d^4$ $3d^5$ $3d^5$ $3d^6$
 $\text{Mn}^{2+} > \text{Fe}^{2+} (\text{I.E})$

Spin only Magnetic moment

$$\mu = \sqrt{n(n+2)}$$

$\mu = n$ point something
 $n = \text{no. of unpaired } e^-$

Ex →

$$C\alpha^{2+} \rightarrow d^4, n=4 \quad 4 \dots$$
$$\text{Fe}^{2+} \rightarrow d^6, n=4 \quad 4. \dots$$

Alloy formation → Alloys are formed by atoms with metallic radii that are within about 15 percent of each other.

E.g. → Brass [Cu (60%) + Zn (40%)]
German silver [Cu (1) + Zn (1) + Ni (2)]
Bronze (Cu + Sn)

Interstitial compound formation

- Prepared by trapping of small sized non-metal like **B, H, C, N** in interstitial space of crystal lattice of metals.
- Non-stoichiometric compound. (VH_{0.56}), (TiH_{1.7})
- Neither Ionic nor covalent
- Hardness ↑ (NiB → hardness similar to diamond) ∴ due to ↑ in packing efficiency
- M.P ↑
- Conductance remain same.
- Chemically inert.

Melting point

M.P depends → (i) unpaired e⁻ in (n-1)d subshell on
(ii) Atomic mass
(iii) Packing efficiency in solid.

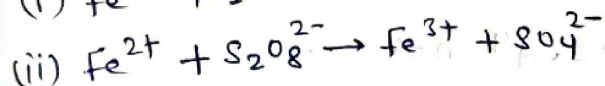
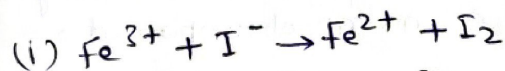
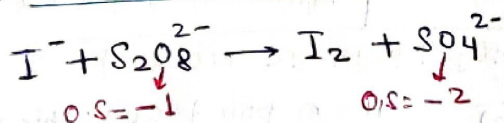
- Max^m M.P in d-block → 'W'
- M.P = Zn > Cd > Hg & Sc > Y > La
Cu > Au > Ag

grp no. → 3rd, 12th 4th to 10th 11th

3d v 4d v 5d	3d ^ 4d ^ 5d	3d v 5d v 4d
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Catalytic properties

(i) Variable oxidation state



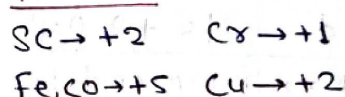
(ii) Complex formation property

- $\text{N}_2 + 3\text{H}_2 \xrightarrow{\text{Fe (Powder)}} 2\text{NH}_3$ (Haber's process)
- $\text{SO}_2 + \frac{1}{2}\text{O}_2 \xrightarrow{\text{V}_2\text{O}_5} \text{SO}_3$ (Contact process)
- veg. oil $\xrightarrow[\text{H}_2]{\text{Ni/Pd}}$ veg. ghee
- $\text{CH}\equiv\text{CH} \xrightarrow{\text{PdCl}_2} \text{CH}_3\text{CHO}$ (Wacker's process)
- Zeigler Natta [TiCl₄ + Al(C₂H₅)₃]
↳ (Use in polymerization)
- Wilkinson catalyst [(PPh₃)₃RhCl]
↳ (Hydrogenation of Alkene)

Variable oxidation state

↳ Due to low energy gap b/w ns & (n-1)d.

NOTE: -



→ Not exist

→ Cu²⁺ is stable & exist only in aq. med.
→ Sc only +3.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
+3	+4	+5	+3	+2	+2	+2	+2	+2	+2
			+6	+7	+3	+3			
			↑ se			↓ se			

ns (n-1)d
↓ ↓
e⁻ of ns = 4 unpaired e⁻ of (n-1)d = x

O.S range = 4 to (x+4)



x = 5, y = 2

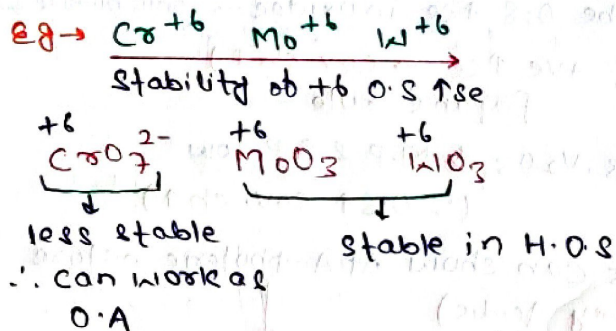
O.S range = +2 to (5+2)
= +2 to +7.

↳ The lesser no. of O.S at the extreme ends stems from either too few e⁻ to lose or share (Sc, Ti) or too many d e⁻ (hence fewer orbitals available in which to share e⁻ with others) for higher valence (Cu, Zn). Thus, early in the series Scandium (III) is virtually unknown & Titanium (IV) is more stable than Ti (III) or Ti (II). At the other end, the only O.S of Zinc is +2. (No d e⁻ are involved).

→ An interesting feature in the variability of O.S of the d-block elements is noticed among the groups (groups 4 through 10). Although in the p-block the lower O.S are favoured by the heavier members (due to inert pair effect), the

Example → In group 6, Mo(VI) & W(VI) are found to be more stable than Cr(VI) . Thus Cr(VI) in the form of dichromate in acidic medium is a strong oxidising agent, whereas MOO_3 & WO_3 are not.

NOTE → In d-block, $T \rightarrow B$ in a group, stability of H.O.S ↑ (Inverse of inert pair effect p-block).



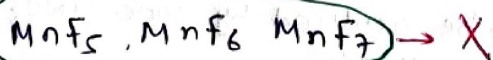
→ Low O.S are found when a complex compound has ligands capable of π -acceptor character in addition to the σ -bonding.

Example → In Ni(CO)_4 & Fe(CO)_5 the O.S of nickel & iron is zero.

Trends in stability of higher O.S

① In case of halides

→ The highest O.S are achieved in TiX_4 (tetrahalides), VF_5 & CrF_6 . The +7 state for Mn is not represented in simple halides but MnO_3F is known & beyond Mn, no metal has a trihalide except FeX_3 & CoF_3 .



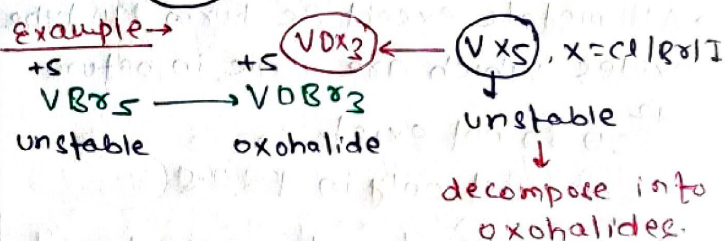
→ due to steric hindrance

→ The ability of fluorine to stabilise the highest O.S is due to either higher lattice energy as in the case of CoF_3 , or higher bond enthalpy term for the higher covalent. eg → VF_5 & CrF_6 .

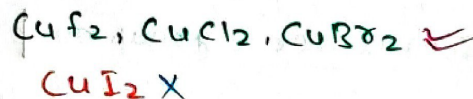
→ Although V^{+5} is represented only by VF_2 , the other halides, however, undergo hydrolysis to give oxahalides VOX_3 . Another feature of fluorides is their instability in the low O.S. eg → VX_2 ($X = \text{Cl, Br or I}$)



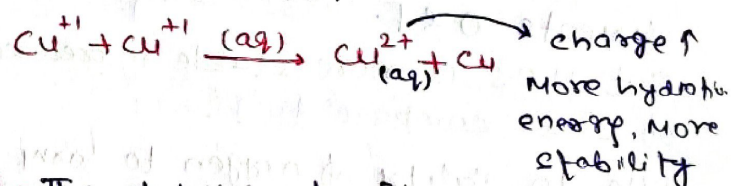
Example →



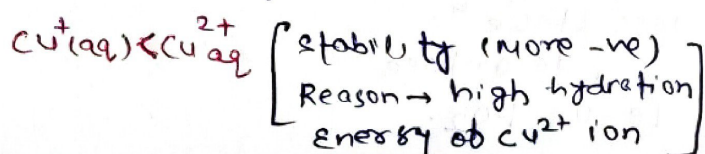
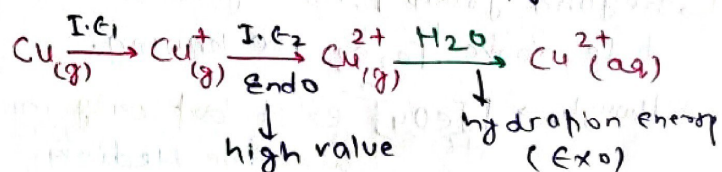
→ Same applies to CuX . On the other hand, all Cu(I) (Cu^{2+}) halides are known except the iodide. In this case, Cu^{2+} oxidises I^- to I_2 .



→ However, many copper(I) comp. are unstable in ('aqueous soln') & undergo disproportionation.



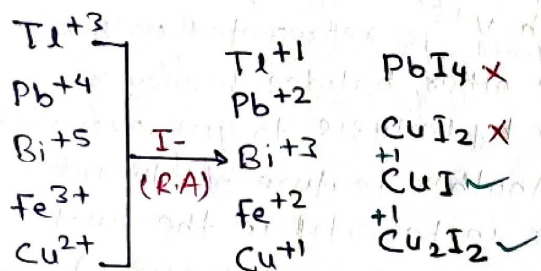
→ The stability of $\text{Cu}^{2+}(\text{aq})$ rather than $\text{Cu}^+(\text{aq})$ is due to the much more negative $\Delta_{\text{hyd}} H^\circ$ of $\text{Cu}^{2+}(\text{aq})$ than Cu^+ which more than compensates for the second ionisation enthalpy of Cu.



In Aqueous Medium



Hydration $\uparrow$ Release energy $\uparrow$ stability $\uparrow$



② In case of Oxides

→ All metals except 'Sc' form M₂O type oxides which are ionic in nature.

ScO do not exist

Sc exist only in +3 O.S

→ The ability of oxide oxygen to stabilise the highest oxidation state is demonstrated in the oxides.

From the group No. 3 to 7
Sc to Mn

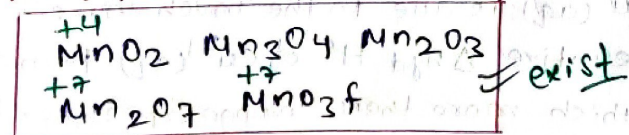
H.O.S of oxides = group no.

NOTE

→ H.O.S of d-Block Metals can be stabilize in presence of High EN elements O & F.

But H.O.S is more stable in presence of 'O' as compare to 'F'.

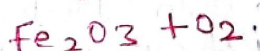
Due to ability of oxygen to form multiple bond with metal.



→ Beyond group 7, no higher oxides of Fe above ⁺³ Fe₂O₃ are known.

Although → [FeO₄]²⁻ exist but only in alkaline medium.

less stable
↓ decompose



→ The ability of oxygen to stabilise these high oxidation states exceeds that of fluorine. Thus the highest Mn fluoride is MnF₄, whereas the highest oxide is Mn₂O₇.

→ The ability of oxygen to form multiple bonds to metals explains its superiority. In the covalent oxide Mn₂O₇, each Mn is tetrahedrally surrounded by O's including a Mn-O-Mn bridge. The tetrahedral [MO₄]ⁿ⁻ ions are known for V(V), Cr(VI), Mn(V), Mn(VI) & Mn(VII).

★ Few More important Point about Oxides

• As the O.S ↑ in oxides → Ionic nature ↓
(∵ +ve ↑ → cov. ch ↑)
Fajan's rule

• CrO₃, V₂O₅ ⇒ M.P & B.P low

(∵ O.S ↑ cov. ch ↑)

• V₂O₅ can show Amphoteric nature (Mainly Acidic)

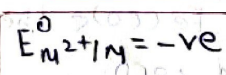
Can react with Both

Base & Acid

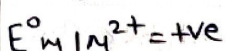
gives VO₃³⁻

gives VO₄⁺

Standard Reduction potential



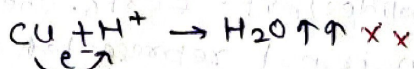
or



→ for all 3-d series

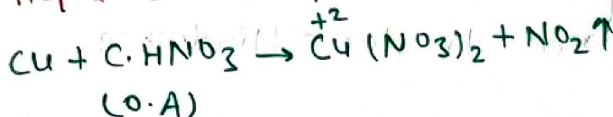
except $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +ve$

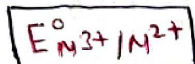
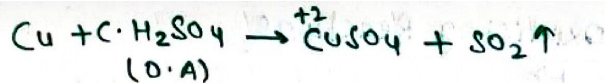
★ That's why Cu can't liberate H₂ gas from acids.



$E^\circ = +ve$ } spont. $E^\circ = -ve$ } Non-spont process.
 $\Delta G = -ve$ } $\Delta G = +ve$ }

→ Cu can be oxidised by conc. HNO₃ & Hot & conc. H₂SO₄.





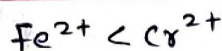
$$E^\circ_{\text{Cr}^{3+}/\text{Cr}^{2+}} = -ve (-0.41)$$

$$E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = +ve (0.79)$$

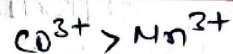
$$E^\circ_{\text{Co}^{3+}/\text{Co}^{2+}} = +ve (1.97)$$

$$E^\circ_{\text{Mn}^{3+}/\text{Mn}^{2+}} = +ve (1.59)$$

Better R.A?



Better O.A?



Imp. points given in NCERT for $E^\circ_{\text{M}^{3+}/\text{M}^{2+}}$ values

① $E^\circ_{\text{Sc}^{3+}/\text{Sc}^{2+}} \rightarrow$ Low value (more -ve)

$\therefore \text{Sc}^{3+} \rightarrow$ Noble gas configuration

② $E^\circ_{\text{Zn}^{3+}/\text{Zn}^{2+}} \rightarrow$ High value [less -ve or +ve]

$\therefore \text{Zn}^{2+} \rightarrow d^{10}$ stable configuration

③ $E^\circ_{\text{Mn}^{3+}/\text{Mn}^{2+}} \rightarrow$ High value (+ve, +1.37)

$\therefore \text{Mn}^{2+} \rightarrow d^5$ half-filled stable config.

④ $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} \rightarrow$ comparatively low

d^5 (less +ve) (+0.79)

$\therefore \text{Fe}^{3+} \rightarrow d^5$ stable config.

⑤ $E^\circ_{\text{V}^{3+}/\text{V}^{2+}} \rightarrow$ comparatively low (-0.26)

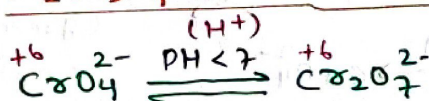
$\therefore \text{V}^{2+} \rightarrow t_{2g}^3$ Half-filled stability d^3

values of $E^\circ_{\text{M}^{3+}/\text{M}^{2+}}$ show that

Mn^{3+}
 Co^{3+} } strong O.A in aq. medium

Ti^{2+}
 V^{2+}
 Cr^{2+} } - strong R.A
- can liberate H_2 gas from acids.

$\text{K}_2\text{Cr}_2\text{O}_7$ (Potassium dichromate)



• chromate ion $\text{PH} > 7$ • dichromate ion $\text{PH} < 7$

• yellow

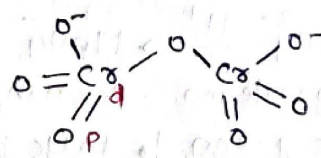
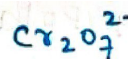
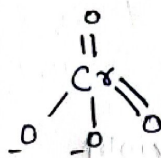
• orange

• stable in

• stable in

Basic med.

Acidic med.



• Avg Cr-O B.O $\Rightarrow \frac{6}{4}$

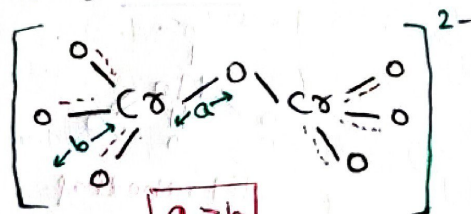
• Two $d\pi-d\pi$

• All Cr-O B.L are equal, only one type of Cr-O B.L are present.

• Four $d\pi-d\pi$

• All Cr-O B.L are not equal.

• Two types of B.L are present.



"b is smaller due to partial double bond character (Resonance)"



• Can be used as a primary standard.

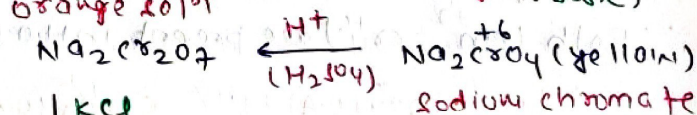
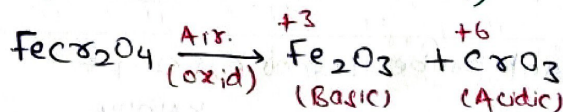
• Can not used as a primary standard in volumetric Analysis.

• More soluble

• Hygroscopic in nature.

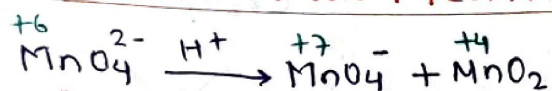
Preparation of $\text{K}_2\text{Cr}_2\text{O}_7$

By chromite ore (FeCr_2O_4) [$\text{FeO} + \text{Cr}_2\text{O}_3$]



$\text{K}_2\text{Cr}_2\text{O}_7 \rightarrow$ Early crystallise due to orange crystals low Ksp (solubility)

KMnO_4 Potassium Permanganate



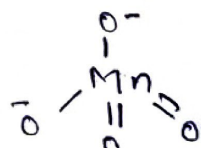
Manganate ion

permanganate ion

- dark green
- stable in basic med.
- $\text{Mn}^{+6} \Rightarrow d^1$
 $n=1$

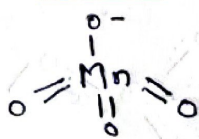
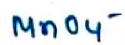
- dark purple or violet
- $\text{Mn}^{+7} = d^0, n=0$
- diamagnetic (But Temp. dependent paramagnetic)

Paramagnetic



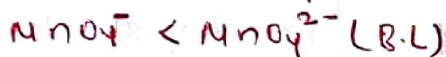
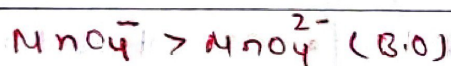
Tetrahedral

- Two $d\pi - p\pi$ Bonds (Mn) (O)
- Avg. Mn-O B.O $\Rightarrow \frac{6}{4}$
- All Mn-O B.L are equal

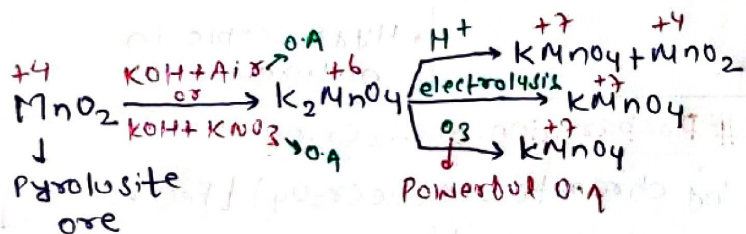


Tetrahedral

- 3 $d\pi - p\pi$ bonds (Mn) (O)
- Avg. Mn-O B.O $= \frac{7}{4}$
- All Mn-O B.L are equal

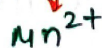
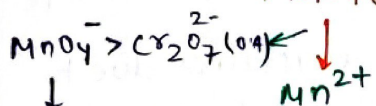
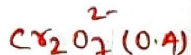
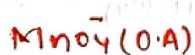


$\Rightarrow$ Preparation of KMnO_4



Important properties of $\text{K}_2\text{Cr}_2\text{O}_7$ & KMnO_4

- Oxidising power $\rightarrow$ Both are strong O.A.
- Both (Mn^{+7} , Cr^{+6}) are present in H.O.S.



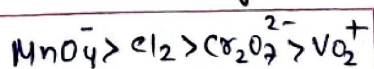
$\therefore$ get easily reduced

to achieve d^5

Half-filled stable

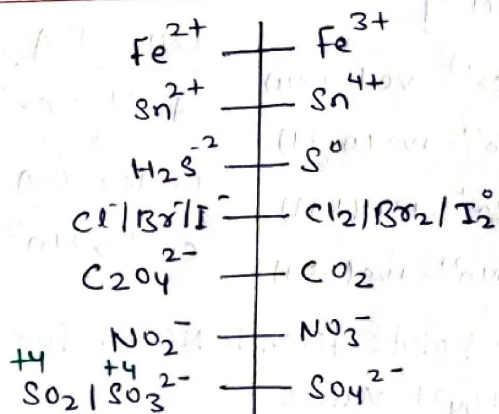
configuration.

(Oxidising power)



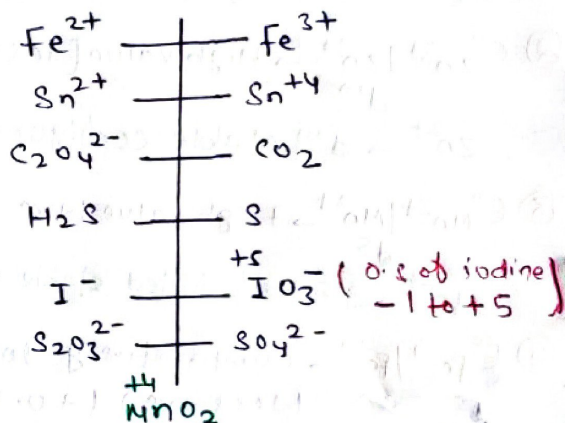
Oxidising Reactions

$^{+7}$ (O.A) MnO_4^- (Acidic, dark purple)

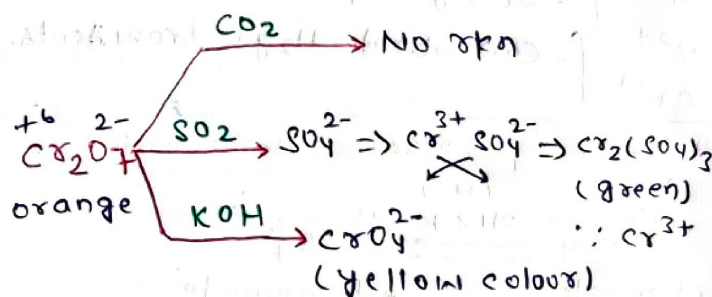
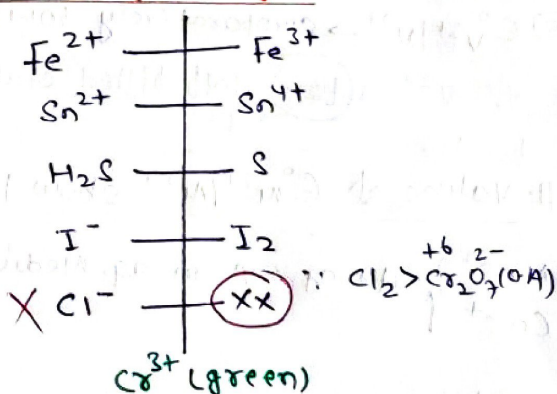


Mn^{2+} (colourless)

$^{+7}$ MnO_4^- (neutral or weak Basic)



$\text{Cr}_2\text{O}_7^{2-}$ (orange)



f-block

Lanthanoids - 58 to 71, $n=6$, group = 3

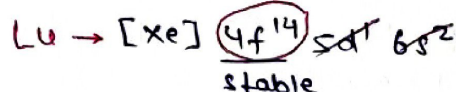
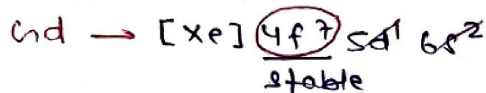
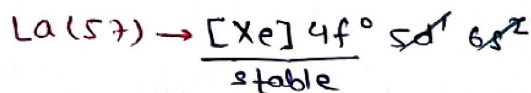
Actinoids - 90 to 103, $n=7$, group = 3

Ionisation Energy

* Lanthanoids :-

- I.E₁ & I.E₂ are 600 kJ/mol & 1200 kJ/mol respectively which is similar to Ca.
- I.E₃ of La, Gd, Lu is lower than expectation as compare to I.E₃ of other elements.

Reason → In teeno mai 3rd e⁻ remove hone ke baad stable configuration achive hoga → Means 3rd e⁻ jaldi dega. ∴ I.E₃ low.



Comparison b/w Actinoids & Lanthanoids

- Size → Lanthanoids < Actinoids
- I.E → Lan. > Act.

∴ I.E ∝ $\frac{1}{\text{Reactivity of Metal}}$ Actinoids are more reactive

- E.N → Actinoids < Lanthanoids (∵ size ↑)

∴ EN ↓ Basic nature of OH⁻ ↑ i.e.
∴ OH⁻ of Actinoids are Basic than Lanthanoids.

Oxidation states

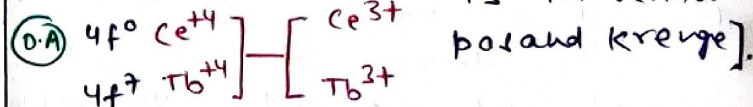
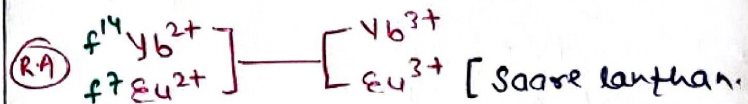
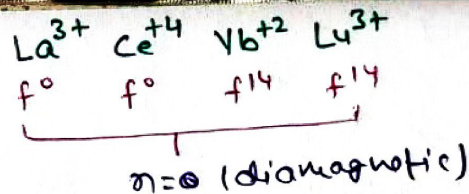
Lanthanoids

- Common O.S = +3
- Although, Eu²⁺, Yb²⁺, Ce⁴⁺ can be shown.
- All trivalent ion of Lanth. are coloured except La³⁺, Lu³⁺

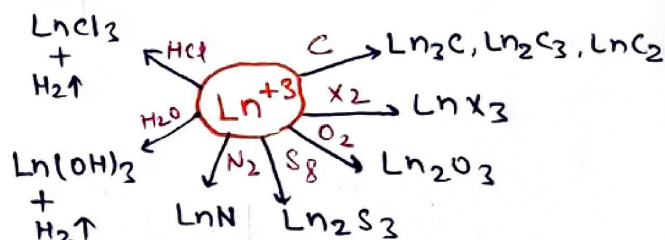
LaLu

Actinoids

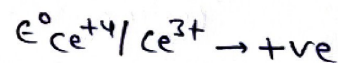
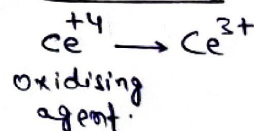
- Common O.S = +3
- Max O.S → Np, Pu (+7)
- But can show variable O.S.
- Due to lesser energy gap b/w 5f & 6d subshells.



* Some Important points (Ln → Lanthano.)



⇒ f⁰ stable



→ Ce⁴⁺ can be used as analytical reagent in aqueous medium.

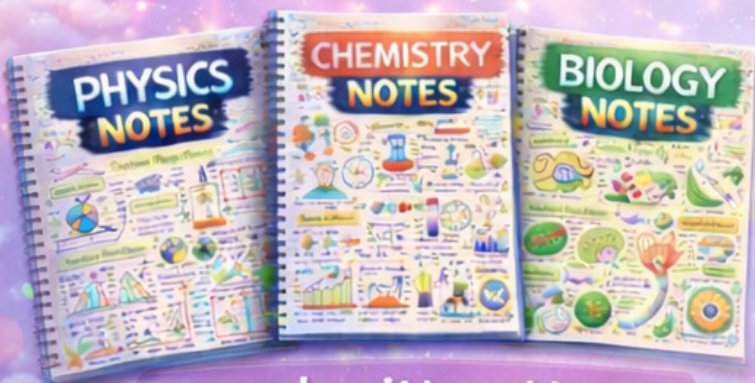
Reason → kinetically rate of rxn is very slow b/w Ce⁴⁺ & H₂O.

Super gift

- Nd → Max. paramagnetic
- Sm → steel like
- Pm → Radio Active
- Misc metal (Metal alloy)
- (Ln = 95%, Fe = 5%, C, S, Ca, Al)
- Ruby (Red) → Cr³⁺ in Al₂O₃
- Emerald (green) → Cr³⁺ in Beryl.
- 'Ulc copper' → copper coated steel.
- Silver Ulc → Cu/Ni alloy.

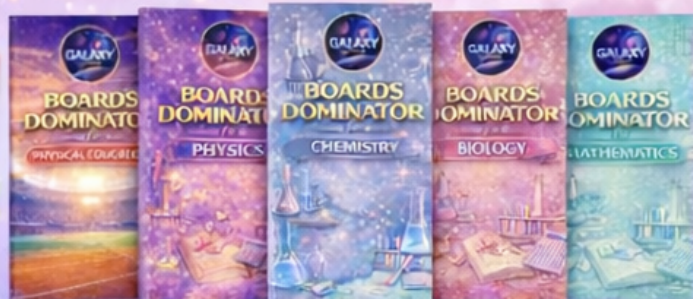
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